



Water-energy nexus in African power pools – The Dispa-SET Africa model

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ABSTRACT

The operation and economic profitability of modern energy systems is constrained by the availability of renewable energy and water resources. Lower water availability due to climate change, higher demand and increased water consumption for non-energy and energy needs may cause problems in Africa. In most African power systems, hydropower is a dominant renewable energy resource, and interconnection capacities are usually limited or unreliable. This paper describes a new modelling framework for analysing the water-energy nexus in the African Power Pools. This framework includes soft linking between two models: the LISFLOOD model is used to generate hydrological inputs and the Dispa-SET model is used for mid-term hydrothermal coordination and optimal unit commitment and power dispatch over the whole African continent. The results show a good agreement between the model outputs and the historical values, despite data-related limitations. Furthermore, the simulations provide hourly time series of electricity generation at the plant level in a robust way. It appears that some African power pools heavily rely on the availability of freshwater resources, while others are less dependent. In the long term, the dependence of the power system on water resources is likely to increase to meet the increasing electricity demand in Africa.

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1. Introduction

Access to a stable and secure supply of energy is a fundamental driver of economic growth in Africa. Around 580 million people, mostly from Sub-Saharan Africa, lacked access to electricity in 2018 [1], and 900 million people had no access to clean cooking facilities such as natural gas, liquefied petroleum gas, electricity and biogas, or improved biomass cook stoves [2]. Africa's economic activity continues its growth. The economic growth is set to reach the 3.4% in 2021 [3]. A key point to support this growing demand is the presence of reliable and effective electricity infrastructure. To this end five regional power pools (regional cooperation entities aimed

to developing a common power grid and a common market between their members) have been established in the last 20 years to improve generation capacity and allow cross-border trade.

Extracting, delivering, and disposing water requires energy, and similarly, many processes for extracting and refining various fuel sources and producing electricity use water [4]. This connection is called 'water-energy nexus' and it is important to understand due to increasing energy demands and decreasing freshwater supplies in many regions. The combined effect of increased water consumption, for energy and non-energy purposes, with lower availability of water resources due to climate change, is expected to lead to monetary losses, power curtailments, temporary shut-downs and demand restrictions in power grids across the world [5]. Similar impacts of climate change on stability of power grid are assessed in Ref. [6] and vulnerability of electricity supply in US and European grids was analysed in Refs. [7,8]. According to Greenpeace [9], global coal industry poses a significant threat to water security in many parts of the world. The world's 8359 coal-fired power

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plants consume 19 billion m³ of freshwater annually, which is enough water to meet the most basic needs of more than 1 billion people. Thus, solutions to the global water security and scarcity crisis must involve water–energy nexus approaches for harmonizing policies and strategies that deal with these inter-related sectors with a main goal of minimizing the environmental impact and maximizing the human benefits [10].

From a modelling perspective, several approaches with different levels of complexity are identified. The small scale economic dispatch of coupled desalinated water, power grid and compressed air storage, formulated as a mixed integer nonlinear programming (MINLP) problem was assessed in Ref. [11]. A similar MINLP formulation was proposed for the analysis of optimal self-scheduling strategies of combined portable water and power generating units [12]. An aggregated joint optimization of regional water–power systems using stochastic dual dynamic programming (SDDP) and applying it to Iberian Peninsula was introduced in Ref. [13]. A more computationally-robust mixed integer linear programming (MILP) optimization of regional water–power systems under cooling constraints and climate change was applied on Spanish [14] and Greek [5] case studies. The synergistic role of renewable energy integration into the unit commitment of the energy–water nexus in Singapore and the Middle East was assessed in Ref. [15]. An “ensemble” of indicators representing water usage spanning different electricity-generation technologies in an attempt to depict the level or lack of detail in current large-scale (Europe) energy-sector water-usage data was assembled and solved with linear programming (LP) algorithm in Ref. [16]. However, highly detailed representation of regional and/or continental systems is observed in only limited number of studies, all of which are based on technological, spatial and/or temporal simplifications.

The electricity demand across all African countries is expected to rise dramatically in the next decades and the entire generation capacity is projected to increase ten-fold by 2065. At the same time consumption of water is expected to increase 7% by 2050 and hydropower, in most African energy systems, will continue to play a very important role, increasing its electricity production from 3 to 5 times in most of the power pools [17]. However, today's energy systems in most of the continent are characterized by their small sizes, the low electrification rates, the high shares of oil in the power generation mix, and the lack of significant power and gas interconnections [18]. Many initiatives such as The United Nation's Sustainable Energy for All [19] and Power Africa [20] focus on the sustainable development of African energy systems. Their aim is to electrify some 60 million homes and support the investment of 30 GW of clean power generation in the near future. Despite this, however, there is no coherent ‘by country’ and ‘by region’ set of scenarios besides the ones proposed by Taliotis et al. [21], nor an open energy system analysis platform that may be used to carry out a more detailed investigation of the proposed long term generation expansion scenarios. Furthermore, a review study [22] of the current state of research on the water, energy, and food nexus showed that in Africa only 7% of studies focus on the simultaneous analysis of water-and energy sectors.

The purpose of this study is: a) to examine current and future electricity situation together with the potential of power trading between countries and b) to investigate the relationship between water and energy by analysing the role of hydropower.

This study aims to refine previous studies, including the TEMBA model (The Electricity Model Base for Africa) [23], with a higher temporal resolution (8760 time periods per year) and a more detailed description of the power fleet.

The synergies between the water and energy sectors are assessed through several what-if scenarios of the availability of water for energy purposes in dry and wet seasons. This paper also

identifies areas where grid extensions would be beneficial to the cost-optimal growth of the African electricity supply system. Cross border interconnections with Europe and other African power pools, despite their significant potential for the future [24], are out of the scope of this paper.

The main contributions of this study are the following:

- An open-source unit commitment and optimal dispatch model covering the whole African continent
- A methodology for the analysis of the water-energy nexus in Africa.
- A compilation of diverse data sources used for the integrated modelling of the five African Power Pools.
- The assessment of the hydro potential in the proposed regions through several what-if scenarios with regard to the availability of water for energy purposes in dry and wet seasons.
- The identification of grid bottlenecks and areas where grid extensions would be beneficial for the African electricity supply system.

This paper is organized as follows: Section 2 presents different tools and techniques used within this study, Section 3 provides insights into the African power system, defines scenarios and describes different inputs and data sources used for benchmarking and analysis. Section 4 focuses on results analysis and shows how each power pool reacts to the wide range of climatic conditions. Finally, Section 5 summarises the main findings and conclusions arising from the analysis.

2. Methods

This section describes the different tools, techniques and methods used within this study. The proposed modelling framework heavily relies on publicly available data and information. Fig. 1 provides an overview of this framework of its interrelationships with the input data. It comprises five main elements: data sources, model inputs, pre-processing, simulation, and calibration. The steps and data used for the calibration exercise are also indicated. The historically available input data, in the form of hourly time series, is complemented with the data from other tools and/or models, such as LISFLOOD, and/or annual reports or statistical databases. Because of relatively low data quality and availability at the continental level, some inputs are benchmarked at a country-level with historical data for 2018 retrieved from IEA, IRENA and IHA datasets as indicated on the right side of Fig. 1. Through an iterative process, generator characteristics are calibrated (indicated with the red connections) to reproduce the 2018 context. These inputs include demand projections, costs, installed capacities (aggregated either per fuel or technology type), net transfer capacities (NTC) and energy flow limits between zones, demand profiles and annual generation. Four models are used to fulfil the objectives of this work:

- LISFLOOD model [25]– used for simulation and generation of multiannual hydro profiles;
- Dispa-SET Side Tools package [26] – used as the data translation and calibration model between the publicly available datasets and LISFLOOD outputs, and the Dispa-SET [27] input database;
- The Dispa-SET mid-term hydro-thermal scheduling (MTS) module – used for the pre-allocation of large storage units (horizon length of 365 days), and
- The Dispa-SET unit commitment and power dispatch (UCM) module – used to optimize the short-term (horizon length of 4 days) unit commitment and power dispatch in considered zones (countries).

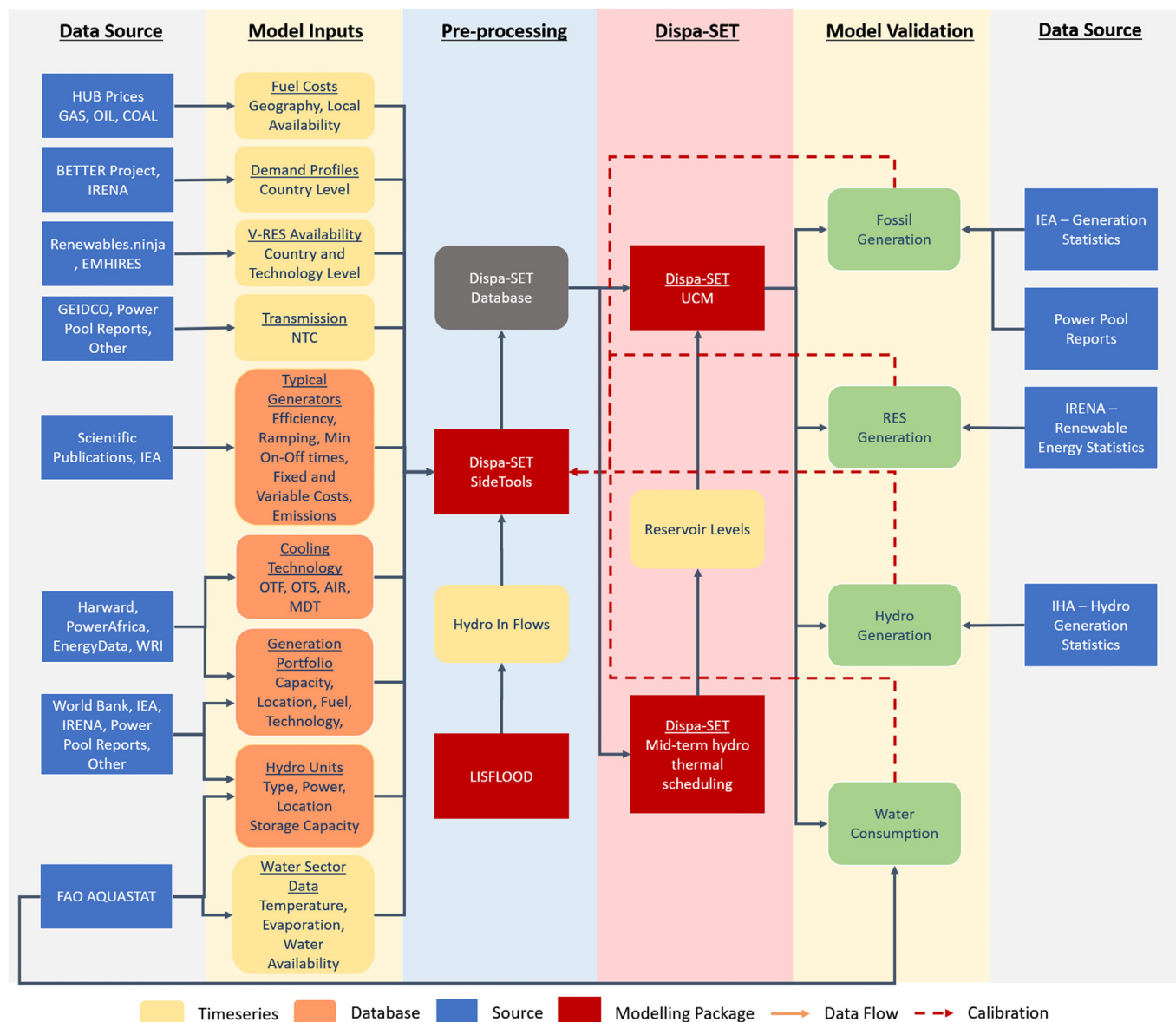


Fig. 1. Flow chart visualizing the different steps for the modelling framework within this study.

2.1. Model inputs and data sources

2.1.1. Data sources

The input data was collected from “best” publicly available sources, some of which were found to be misleading, inadequate or contradictory. To support earlier studies at different levels and facilitate the improved management of water resources, significant effort was dedicated to improving data availability by linking observations and projections of power and water systems at relevant technical, economic, spatial and temporal levels using common sense and assumptions. For example, in some cases, the data was complemented or adjusted using a more detailed dataset (e.g. relative to a single country or to a single river basin). Furthermore, all data is openly available under the FAIR Guiding Principles for scientific data management and stewardship allowing users to modify or recreate the model in other simulation environments [28]. The data processing workflow is also thoroughly documented and is provided as electronic annex of this paper.

2.1.2. Cooling systems assumptions

Despite the use of multiple data sources, more than 30% of the units were left with unidentified cooling technology. In order to fill this gap, a cooling system selection matrix based on different fuel, technology (open cycle gas turbines – GTUR, combined cycle gas turbines – COMC, internal combustion – ICEN and steam turbines – STUR) and geography (proximity to large bodies of water) combinations was applied (Table 1). This matrix is based on previous work from Macknick et al. [29] in the United States, Fernandez-Blanco et al. [5] in the Iberian peninsula, and Pappis et al. [30] for several African countries. The identification process was carried out as follows: aerial distances between each power plant and large body of water (oceans, seas, lakes and big rivers such as Kongo and Nile) was computed. If the power plant is located within the 10 km radius the once-through system was assumed as default cooling technology for all HRD and GAS units. Otherwise, the cooling system allocation matrix was used for assigning one of the three cooling technologies, air cooling (AIR), mechanical or natural draft cooling tower (MDT) or once through cooling using fresh or saline

Table 1

Cooling system allocation algorithm for power plants where cooling type is not known. Values in the table indicate installed power plant capacities (MW).

Fuel	Id	Cooling	GTUR	STUR	ICEN	COMC
Biogas, Biomass, Straw, Wood, etc.	BIO	AIR	<15	<15	<15	<15
Geothermal	GEO	MDT	≥ 15	≥ 15	≥ 15	≥ 15
Hard Coal, Anthracite, Bituminous Coal, Coke, Steam Coal, etc.	HRD	AIR	—	<200	—	—
		MDT	—	200–450	—	—
		OTS	—	≥ 450	—	—
Natural Gas, Butane, Methane, Propane, Liquefied and Synthetic Natural Gas	GAS	AIR	>0	—	<15	200–250
		MDT	—	<20	—	>0
		OTS	—	≥ 20	≥ 15	180–720
Crude Oil, Diesel, Fuel Oil, Jet Fuel, Kerosene, Light Fuel Liquefied Propane Gas, Methanol, Naphtha	OIL	AIR	>0	<15	>0	<15
		MDT	—	≥ 720	—	≥ 720
		OTS	—	15–720	—	15–720
Peat Moss	PEA	MDT	>0	>0	—	—

water (OTS). Fresh water consumption was assigned only to units outside the ± 10 km range from oceans and seas.

2.2. Pre-processing

2.2.1. Computation of hydro dam inflows and run-of-river availability factors

Hydropower inflows in m^3/s are computed using a standard LISFLOOD model [31] setup for a particular river basin and particular geographical location. This setup includes a three-layer simulation of soil-water balance, flow of subsurface and groundwater, routing of surface runoff to the nearest river channel and routing of the channel flow. Besides, snow melt, infiltration, interception of rainfall, leaf drainage, evaporation and water uptake by vegetation, surface runoff, preferential flow (bypass of soil layer), exchange of soil moisture between the two soil layers and drainage to the groundwater, sub-surface and groundwater flow, and flow through river channels are considered [32]. LISFLOOD outputs usually generate relatively high water availabilities (WA) for energy needs only and for the considered location. In order to assess the share of WA for hydropower generation, technological features, such as nominal head, maximum power capacity, volume and surface area of the reservoirs, as well as satellite-based data such as average, minimal and maximal daily air temperatures and daily solar irradiation, need to be properly assessed. Evapotranspiration is calculated for each unit and geographical location, as proposed by Ref. [33]. They are then subtracted from the LISFLOOD outputs.

For some hydro units, the computed inflows (potential energy in accumulation reservoirs behind the dams, in MWh) are orders of magnitude higher than the historical generation. This is due to the limited quality of the input data and due to inaccuracies in the definition of the catchment basins for these units. To remediate this, parameters such as hourly availability factors for hydro run-of-river (HROR) units and capacity factors for reservoir-based hydropower without pumping capability (HDAM) and pumped storage (HPHS) units are adjusted using an iterative two-stage calibration method (Fig. 2).

The first iteration computes the difference between annual generation and computed inflows. Based on the unit type, one of the two scaling methods is selected. For HDAM and HPHS units, inflows are scaled proportionally to match the yearly output. It is assumed that spillage is low and can be neglected thanks to large storage reservoirs usually representing several hundreds of hours of full load capacity. In the case of HROR units, a second method is applied to account for the lost energy due to spillage. This typically occurs when inflows are higher than the design flow rate of the turbine. The following iterative five step methodology is introduced:

$$\text{I) } InFlow_{u,i} = InFlow_{u,i} \cdot (1 + Correction_u)$$

$$\text{II) } Spill_{u,i} = InFlow_{u,i} - PowerCapacity_u$$

$$\text{III) } Spill_{u,i} (Spill_{u,i} < 0) = 0$$

$$\text{IV) } InFlow_{u,i} = InFlow_{u,i} - Spill_{u,i}$$

$$\text{V) } Correction_u = \frac{HistoricGeneration_u - \sum_{i=1}^{i=n} InFlow_{u,i}}{\sum_{i=1}^{i=n} InFlow_{u,i}} \quad (1)$$

these inflows are scaled to match the yearly output, and spillage recomputed using the given inflows. If the newly computed HROR is close to the historical values, the iteration stops, otherwise new correction factors are applied to the initial inflows and the next iteration starts.

2.3. Model validation and outputs

2.3.1. Power system model

The central point of this study is the computation of optimal operation of power generation fleet in several interconnected power pools. This is achieved using the open-source Dispa-SET model, a cross sectoral unit commitment and power dispatch model [34]. As indicated in Fig. 1, optimization in Dispa-SET is split into two stages: a simplified MTS and a classical UCM stage. Both stages use the same input dataset, pre-processed and created by the intermediate Dispa-SET side tools package, as described in the previous sections. The short-term Dispa-SET UCM module optimizes the power dispatch of individual generator units, energy flows between interconnected zones, variable renewable energy source (VRES) curtailment, shed load, carbon emissions, interactions between water and energy sector, shadow prices, water values etc. It is expressed as a MILP problem and integrates diverse clustering techniques and a rolling horizon optimization to ensure computational tractability [35]. The model outputs are then analysed, and relevant water-power nexus metrics are computed. A detailed formulation of the Dispa-SET model is out of the scope of this paper. All relevant equations and constraints can be found in Ref. [36], and detailed description of the clustering approaches is further discussed here [35], objective function and power demand balance, as well as a more detailed description of the Dispa-SET MTS module are provided in the Annex A, located at the end of this paper. The main constraints of the model are as follows:

- Technical limits on minimum and maximum power generation for individual units,

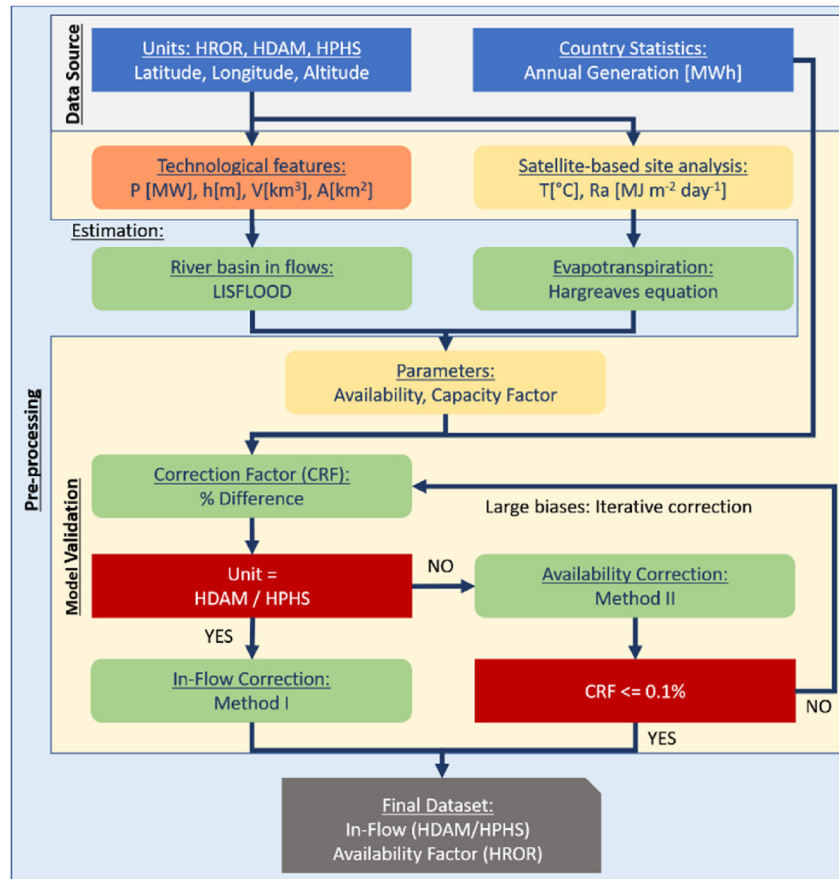


Fig. 2. Flow chart visualizing the generation of availability factors time series for HROR and scaled inflows time series for HDAM units.

- Multiple storage technologies (hydro, thermal, etc.)
- Ramping limits, start-up and shut down times and minimum on and off times,
- Non-dispatchable units (e.g. wind turbines (WTOT), solar photovoltaics (PHOT), HROR, etc.)
- VRES Curtailment,
- Shed load as the main measure of the adequacy of the system,
- Up and down reserve requirements (spinning and non-spinning),
- Detailed representation of fixed and variable costs
- Multiple nodes with constrained interconnection capacities (congestion management)

2.3.2. Power system indicators

The following key power system indicators are computed from the model output:

- **Shadow prices** are computed for each hour and for each zone. The shadow price of electricity is the dual value of the energy balance equation. It can be interpreted as the clearing price of an ideal wholesale market.
- The need for **load shedding** highlights the inadequacy of the system. It is defined as the demand of the system that must be reduced to match the available supply [34]. A maximum value of the load shedding capacity is defined for each simulated country using the generic formula as described in Ref. [37].
- **VRES curtailment** refers to the intentional reduction of renewable generation. The total curtailed energy computed as a

share of total available VRES generation and reflects the lack of flexibility in the system.

- The **carbon footprint** is computed using the standard emission factors for different fuel and technology combinations, as proposed in Ref. [38]. These emissions relate to emissions from power generation and operation of thermal units only (life cycle emissions are not considered) and are disaggregated on per unit level.

2.3.3. Water stress and water exploitation indicators

Evaluating the resilience of the system is important and should include the rational exploitation and utilization of resources, as irreversible changes may occur once the external disturbance beyond a certain range [39]. Thus, the water exploitation index (WEI) is defined as an indicator of water stress, relating water uses (in this study this refers to water needs for energy sector only) to water availability (WA) as proposed by Ref. [40]. Within this study, several indicators of water use are defined:

- **Water withdrawal (WW)** is the abstraction of water for an economic activity. Literature also refers to it as gross water abstraction or withdrawal [29].
- **Water consumption (WC)** is defined as the portion of water not being returned to the original water source after being withdrawn. WC occurs when water evaporates into the atmosphere or diverted for other usages. WC is also commonly referred to as net water abstraction or blue water footprint [29].

- **Absolute water exploitation index** (WEI^{Abs}) refers to the ratio of WW over WA. It can be any positive number, but normal range is between 0 and 5. Values above 1 do not indicate that more water is being withdrawn than potentially available because same water can be withdrawn multiple times along the supply chain (i.e. two hydro dams operating as a cascade on the same river are withdrawing same water two or more times).

$$WEI^{Abs} = \frac{WW}{WA_{Int} + WA_{Ext}} \quad (2)$$

- **Consumptive water exploitation index** (WEI^{+}) stands for WC over WA. It typically ranges between 0 and 1, but values above 1 are also possible in extreme cases. Values above 0.2 are considered as critical in terms of water scarcity.

$$WEI^{+} = \frac{WC}{WA_{Int} + WA_{Ext}} \quad (4)$$

These indicators are computed per country and per power pool.

3. Scenario analysis

The geographical overview of African power pools analysed in this study is presented in Fig. 3. This is an extension of previous works done by the same group of authors, including a two separate models for the West African Power Pool (WAPP) [18], Southern African Power Pool (SAPP) [41], and another model covering the Central, Eastern and North Power Pools [42,43].

In order to evaluate the potential flexibility originating from the water-energy nexus in the African power pools, two scenarios are

defined: a Baseline scenario referring to the power system from year 2018 and a High interconnections scenario referring to the short-term expansion of the African power grid. Each scenario includes a detailed sensitivity analysis in which the power system is simulated over 39 climate years. A climate year consists of a set of meteorological variables, in this study climate year hydrological conditions, that are used to simulate the climate variability in the power system. In other words, all the non-climatic inputs/parameters of the power systems are kept the same while the hydrological conditions (i.e. hydropower inflows) from the climate years are used.

3.1. Baseline scenario

The Baseline scenario corresponds to the year 2018 and is run for the 39 climate years. NTCs between neighbouring power pools and NTCs between individual countries within the power pools are constrained by the existing power grid. Several countries (Central African Republic in CAPP, Eritrea and Somalia in EAPP and Gambia, Guinea, Guinea-Bissau, Liberia, Niger and Sierra Leone in WAPP) that do not share any cross-border lines with the neighbouring countries are simulated in isolation. Although the results from this study are grouped by power pool, the simulations are carried out at the country level. Input data, presented in the upcoming sections, is kept constant for both scenarios.

3.1.1. Hydrology

The availability of water resources for hydropower generation shows a strong inter-annual variability, as presented in Fig. 4. For some regions, the range of the water inflow variations within the year is particularly wide, as for example for CAPP, where the share renewables range from 50.7% (in 1983) to 77.6% (in 1988).

3.1.2. Capacity mix

The total generating capacity of the African power pools sums up to 217.96 GW as shown in Table 2. Most values are based on the year 2018, but due to limited data availability, in some countries, latest known values were used instead. The sources for thermal power plant capacities are the “World Electric Power Plants Database” [44], the “African Energy Portal” [45], the “Global Power Plant Database” [46], Renewable Energy Statistics” [47], IEA [48] and IHA [49].

The largest share of installed generation capacity, around 76% (165.7 GW), corresponds to the thermal power plants (OIL, GAS and HRD). The second largest share of around 16% (35.8 GW) corresponds to hydro generators, and only 5% (11.6 GW) of the total installed capacity corresponds to variable renewables such as wind (WIN) and solar (SUN). The share of other renewable sources, such as bio-based fuels (BIO) and geothermal (GEO), corresponds to only 1% (2.8 GW). Situation across different power pools is however diverse, as shown in Fig. 6.

Three hydro technologies are considered: HDAM totalling 24.6 GW, HPHS totalling 3.4 GW and HROR 7.85 GW. This study also distinguishes two solar power technologies, namely the solar PV (PHOT) totalling 5.1 GW and concentrated solar power (SCSP) totalling 1 GW, and two wind technologies: onshore wind (WTON) and offshore wind (WTOF). Share of installed capacity by power source is presented in Fig. 5.

3.1.3. Storage capacities

There are three main storage technologies: HDAM, HPHS and SCSP. Their capacities are provided in Fig. 6. HDAM units allow the delay of the use of the water to generate electricity later (from hours to months), HPHS gives the possibility to store the surplus of electricity (generally during the low demand periods) and SCSP

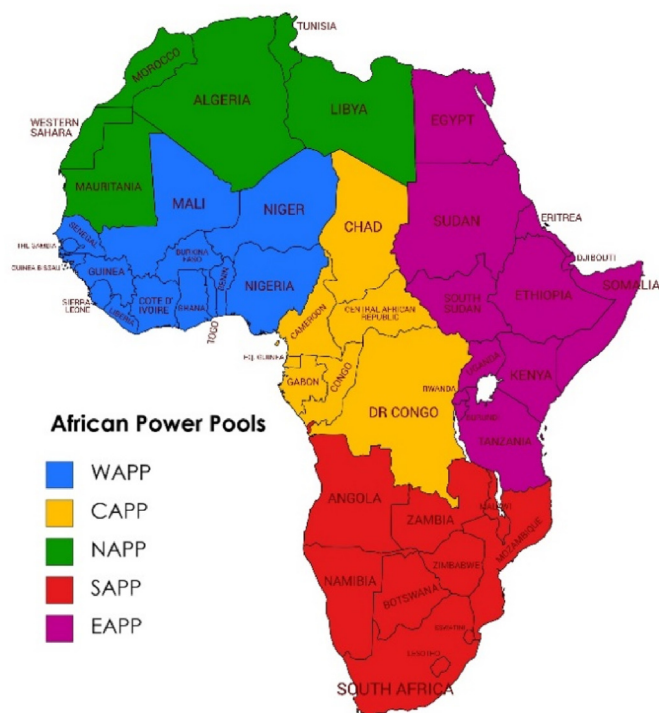


Fig. 3. Classification of African Power Pools (The same colour code is used for individual power pools throughout the paper). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

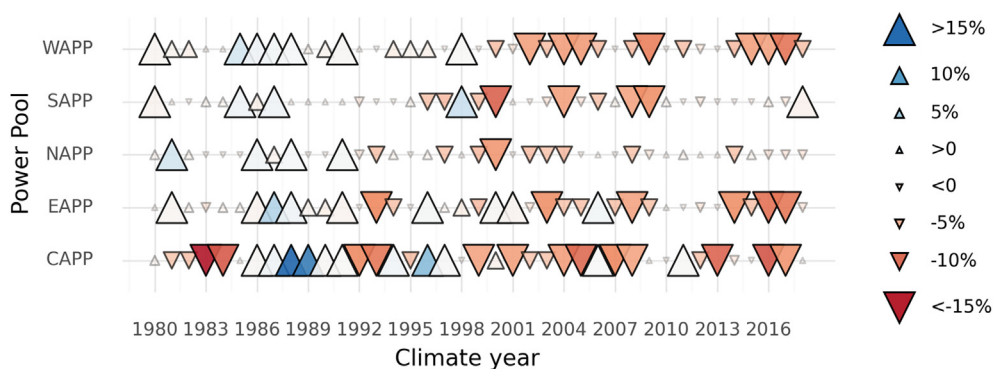


Fig. 4. Annual variability (percentage deviation from the mean) of daily inflow in the 39 climate years for the considered power pools. The size of the triangles indicates the inter-annual variability, the direction of the rectangles highlights the increase or decrease compared to the mean and colour indicates the intra-annual (seasonal) variability within the individual power pools. Simulation results in all climate related diagrams do not reflect changes in generation capacity over time. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 2

Installed capacity in individual power pools and Africa as a whole grouped by the generation type (installed capacities are expressed in GW).

	Total	BIO	GEO	PEA	OIL	GAS	HRD	NUC	WAT	SUN	WIN
CAPP	6.75	<0.01	—	—	0.98	1.43	—	—	4.29	0.04	<0.01
EAPP	74.27	0.93	0.79	0.02	9.05	49.32	—	—	11.17	1.18	1.81
NAPP	46.12	0.01	—	—	5.25	31.34	4.32	—	2.27	1.32	1.61
SAPP	70.67	0.93	—	—	5.59	2.46	42.06	1.93	12.44	3.17	2.10
WAPP	20.16	0.17	—	—	5.09	8.09	0.75	—	5.67	0.40	0.01
Africa	217.96	2.02	0.79	0.02	25.96	92.64	47.12	1.93	35.84	6.11	5.52

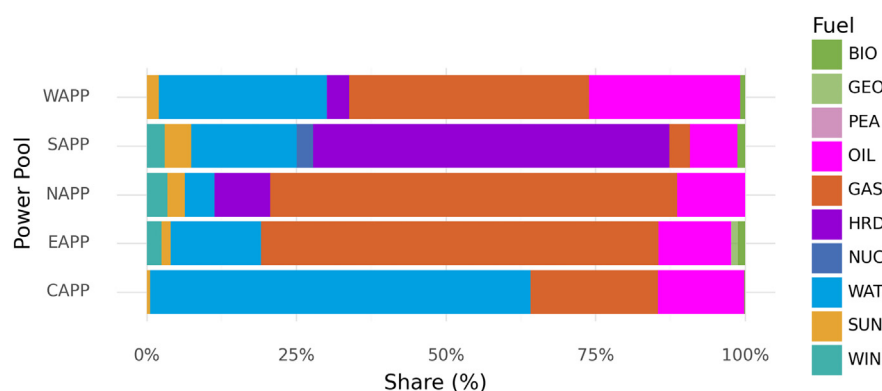


Fig. 5. Share of capacity by power generation type in the analysed power pools.

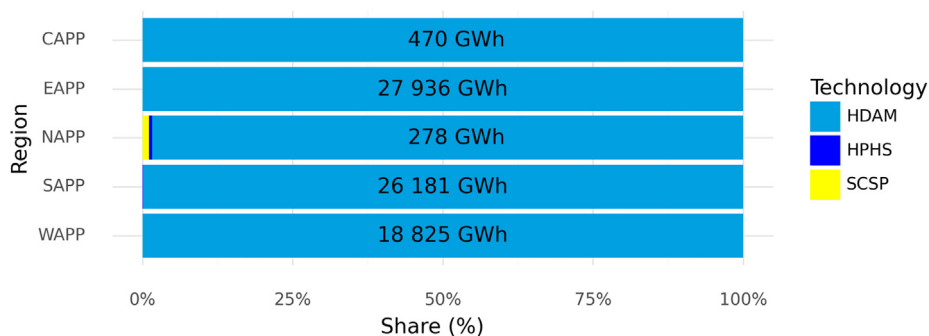


Fig. 6. Maximum amount of storable electricity in each power pool sorted by technology (hydro dams, pumped storage and concentrated solar power).

power plants can usually store energy in the form of sensible heat (in molten salts) with 8–15 h of storage. The main reason for this is the fact that most hydropower plants are either HROR or HDAM with limited storage capacity.

3.1.4. Fuel prices

A summary of the fuel prices is presented in Fig. 7. Variability of the prices across different countries and power pools is based on the fuel price forecast algorithm (ANNEX B), taking into account the influence of resource availability, fuel supply network and geographical location. The marginal price of VRES, geothermal and hydro are assumed to be zero.

3.1.5. Power grid and cross-border interconnections

The existing grid infrastructure in sub-Saharan countries is relatively weak. Countries such as Central African Republic and South Sudan experience system-wide outages for up to 10 h per day [50]. A better situation is observed in NAPP and SAPP, where more than 97% and 87% of the population has access to electricity respectively and national grids are modernized. NAPP and SAPP are also the most interconnected power pools, followed by EAPP, where Eritrea and Somalia [51] do not share any cross-border lines, and CAPP where Central African Republic [52], Chad [53] and most of the Democratic Republic of the Congo are isolated from the main grid. Nevertheless, several interconnection projects such as Kenya-Ethiopia [54], Kenya-Tanzania-Uganda [55], Rwanda and Democratic Republic of the Congo [56] and strengthening of the WAPP power grid [57] are already under construction.

3.1.6. High Interconnections scenario

The High Interconnections scenario is the potential short-term grid expansion “what-if” scenario. The main motivation behind this scenario is to analyse the impact of planned grid expansion projects for the year 2025 on the current system configuration (installed capacities, VRES capacity factors and techno-economical parameters are kept the same as in the Baseline scenario). The input data is obtained from the projects currently in development and from studies and reports from individual power pools and national regulatory agencies. A summary of the reports used for the projecting the short-term grid expansion is presented in Table 3. The vast majority of the cross-border projects mentioned in this study are already in the mid to late stage of the development or are currently being implemented, meaning that their financing is already secured. Thus, analysis of the costs associated to the short-term expansion of the grid is out of the scope of this paper. Total existing and planned NTC's between different power pools are

shown in Fig. 8. The largest increase of NTC's is expected between EAPP and the Arabic states (x6) and NAPP and the EU (x5), two regions excluded from this analysis. Largest increase between analysed power pools is expected between CAPP and WAPP (x330, significant increase due to establishment of the first ever cross-border capacity) CAPP and EAPP (x4) EAPP and SAPP (x3.5) and EAPP and NAPP (x3). Thus the highest increase of cross-border transmission capacities is surrounding the EAPP, only power pool that could exchange electricity with at least three other ones.

4. Results

This section shows how each power pool reacts to the wide range of climate conditions (mainly hydropower inflows). The simulations from both scenarios, and over the 39 climate years, produce outputs in the form of an hourly time series describing the state of the power system (commitment status of individual units or clusters of units, costs of dispatching them etc.).

4.1. Energy mix

The simulation results indicate that the share of renewable power generation varies between 14.8% (climate year 2017) and 21.9% (climate year 1987). The average annual generation from renewable sources is 156.3 TWh (deviation range is between 81% and 110%). Fig. 9 shows the energy mix for the 39 climate years. Among the analysed power pools, the CAPP shows the widest variation, between 50.7% and 77.6%. In most power pools the largest portion of variability is caused by hydropower, while in NAPP, it is caused by wind and solar generation. In the High interconnections scenario minimum VRES generation (dry weather year) on the continental scale increases from 14.8% to 16.2%, while maximum VRES generation increases from 21.9% to 22.1%. In individual power pools, largest increase in VRES generation is observed in CAPP, where up to 93.7% (an increase of 16.1% points) of electricity could be generated by VRES units.

4.2. Hydro generation

On average, about one fifth of the electricity comes from the hydropower. Fig. 10 illustrates the hydropower generation in all the power pools and different climate years. For run-of-river technologies the largest variability (normalised standard deviation of 27%) is observed in CAPP while hydro storage technologies, the largest variability (normalised standard deviation of 29%) is observed in SAPP. Total generation from storage and pumping units at the

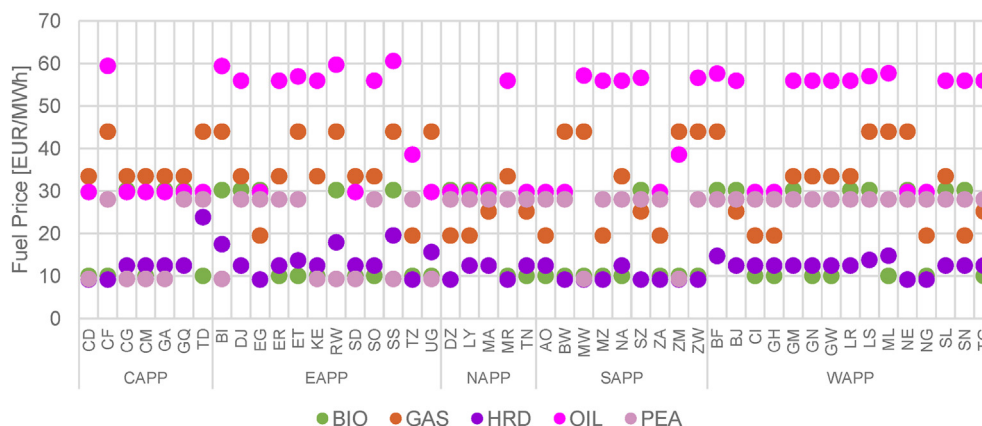


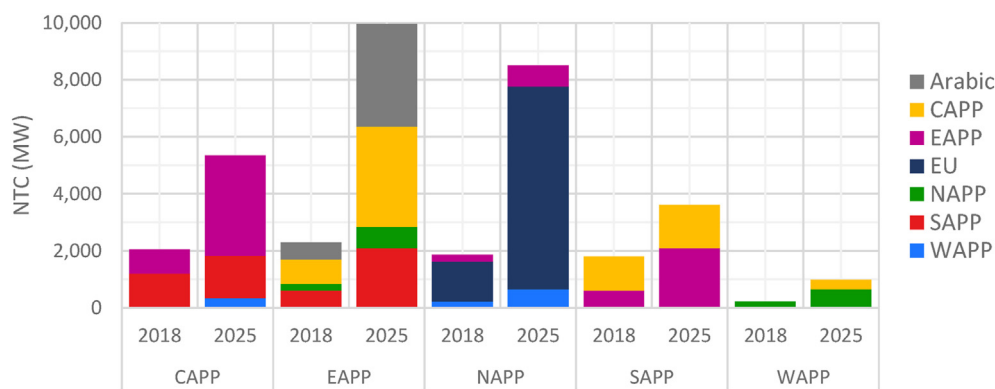
Fig. 7. Summary of the fuel prices across all analysed countries (two letter country codes – ISO2) and all five power pools.

Table 3

Table of sources used for projecting the short-term grid expansion.

Source	Coverage					Comment
	CAPP	EAPP	NAPP	SAPP	WAPP	
[21,23]	M	M	M	M	M	Basic data on energy and capacity mix, VRES and power grid infrastructure
[57]	-	-	-	-	H	Extensive data on power grid infrastructure, energy and capacity mix
[58]	H	H	H	H	H	Extensive data on fossil reserves, energy and capacity mix, VRES and power grid infrastructure
[59]	-	M	-	M	-	Extensive data on power grid infrastructure
[60]	L	M	-	L	-	Extensive data on power grid infrastructure
[61]	H	-	-	L	-	Extensive data on energy and capacity mix, VRES and power grid infrastructure
[62]	-	L	H	-	-	Extensive data on power grid infrastructure
[63]	-	-	-	H	-	Extensive data on power grid infrastructure

Level of detail: L - Low, M - Medium, H - High.

**Fig. 8.** Total NTC (stacked columns) and NTC among the power pools (colour). Year 2018 represents historical NTCs, while year 2025 represents planned short-term grid expansion according to existing grid extension plans. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

continental scale exceeds that of run-of-river units by a factor of five. CAPP is the only power pool where run-of-river is the dominant hydro technology.

4.3. CO₂ emissions and generation costs

The variability of renewable energy generation (at the continental scale, hydro contributes to more than 85% of total VRES generation), also impacts CO₂ emissions and generation costs. The observed range for the CO₂ intensity ranges between 434 and 473 kgCO₂/MWh for climate years 1987 and 2017 in the Baseline scenario (Fig. 11). The largest CO₂ intensity is observed in SAPP (589–659 kgCO₂/MWh in Baseline and 596–653 kgCO₂/MWh in High Interconnections scenario) and the lowest in CAPP (139–303 kgCO₂/MWh in Baseline and 38–228 kgCO₂/MWh in High interconnections scenario). The average generation cost on the continental scale ranges between 23.23 and 25.69 EUR/MWh for climate years 1987 and 2000 in the Baseline and 22.03–24.09 EUR/MWh for climate years 1987 and 2017 in the High interconnections scenario. The impact of the improved cross-border interconnections on the reduction of CO₂ intensity and generation cost in the High Interconnections scenario are best observed in CAPP and WAPP. However, new NTC additions between NAPP and WAPP and NAPP and EAPP barely impact CO₂ intensity and generation cost in NAPP. For a more uniform price and lower CO₂ intensity across all power pools, significant increase of cross border capacities would be required.

4.4. Power system adequacy

4.4.1. Curtailment and shed load

Load shedding and curtailment can be considered as good indicators of system adequacy and flexibility, respectively. At the continental scale, the median curtailment from the 39 climate years equals 4.49 TWh in Baseline and 3.08 TWh in the High interconnections scenario. The largest curtailment levels are observed in CAPP (up to 30% in both scenarios) while there is almost no curtailment in NAPP. The need to shed load is observed in all power pools at the exception of SAPP. In the Baseline scenario the supply and demand balance cannot be guaranteed in Central African Republic, Gabon, South Sudan, Somalia, Guinea Bissau, Democratic Republic of Congo and Libya. The High Interconnections scenario does, however, resolve the unbalance and suppresses most of the observed shed load. The dependency of VRES curtailment and load shedding versus power output from hydro and fossil units is shown in Fig. 12.

4.4.2. Power feed and congestion

The energy produced in each zone and transfers between them are presented in Fig. 13. In both scenarios South Africa, Zambia, Zimbabwe, Ethiopia, Kenya, Nigeria, and Ghana have important capacities and are net exporters. It is noteworthy that the lack of intra-regional cross border interconnections, especially in WAPP and CAPP, is significant cause of suboptimal dispatch.

For example, Guinea does not share any interconnections with

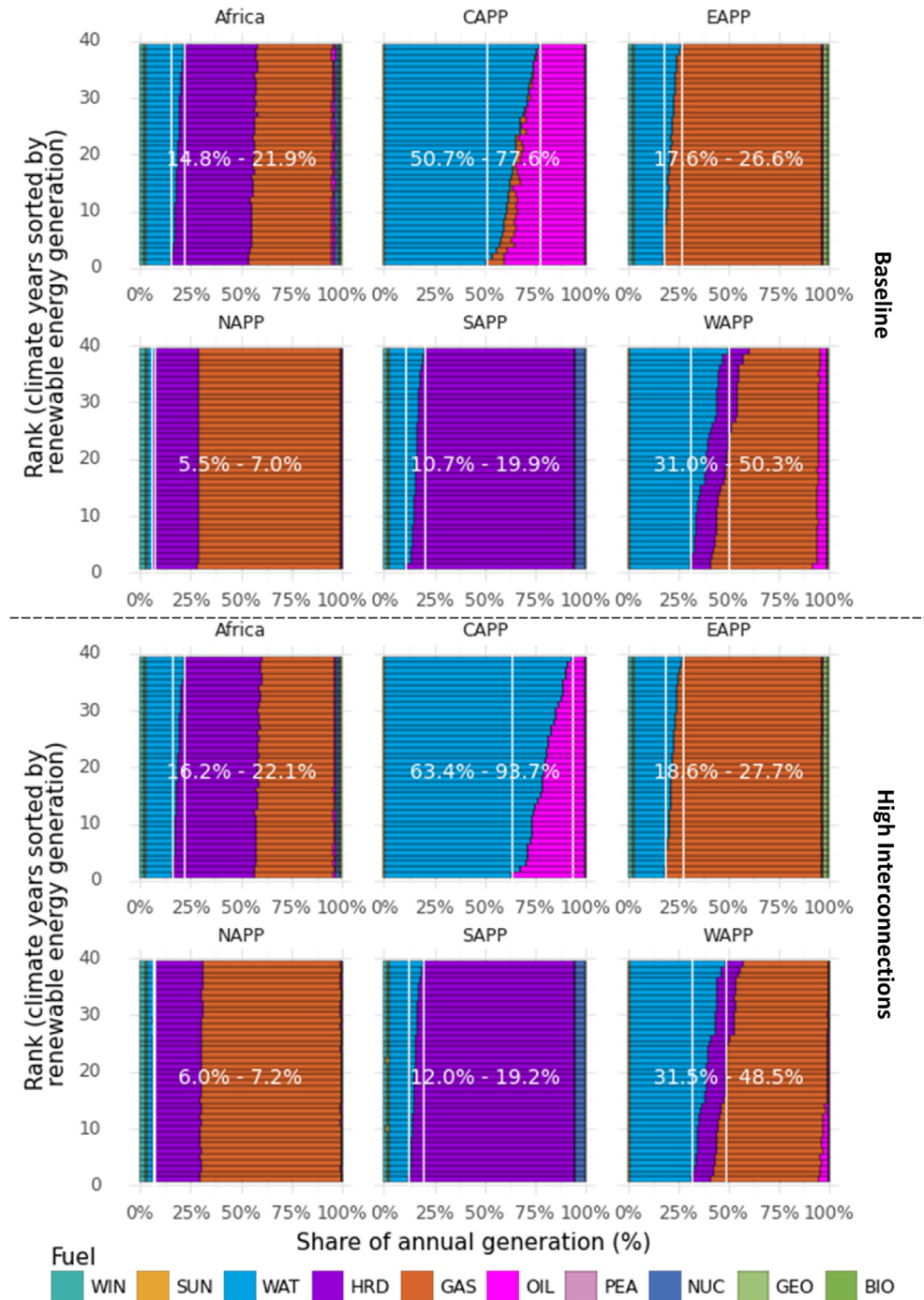


Fig. 9. Generation mix and share of renewables for the considered climate years and both scenarios. White lines indicate minimum and maximum shares of renewable generation in individual power pools and Africa as a whole. Climate years are ordered by the share of renewable generation.

its southern neighbours which leads to an underuse of hydro dams balancing potential in the region. In the High interconnections scenario, new cross border capacities allow unlocking part of this potential to more uniform shadow prices, reduced VRES curtailment and load shedding and lower CO₂ emissions. Limited interconnections in the Baseline scenario result in highly congested lines from SAPP to Tanzania. This is only marginally reduced in the High interconnections scenario, despite the new NTC additions. The cross-border congestion is presented in Fig. 14. Overall, in the fully interconnected grid, flows from north to south and to some extent

from east to west are made possible.

Fig. 15 shows the range of average weekly net power flows in individual power pools. In the Baseline scenario SAPP is the only African power pool that exports electricity in all the 39 climate years. In the High interconnections scenario, the net exports of SAPP are even higher. The main difference is observed in EAPP, which becomes a net exporter. The highest congestion in the Baseline scenario is observed between EAPP and NAPP (97% of the time), and EAPP and SAPP (99%). In the High interconnections scenario, congestions between EAPP and NAPP (92%), and EAPP and

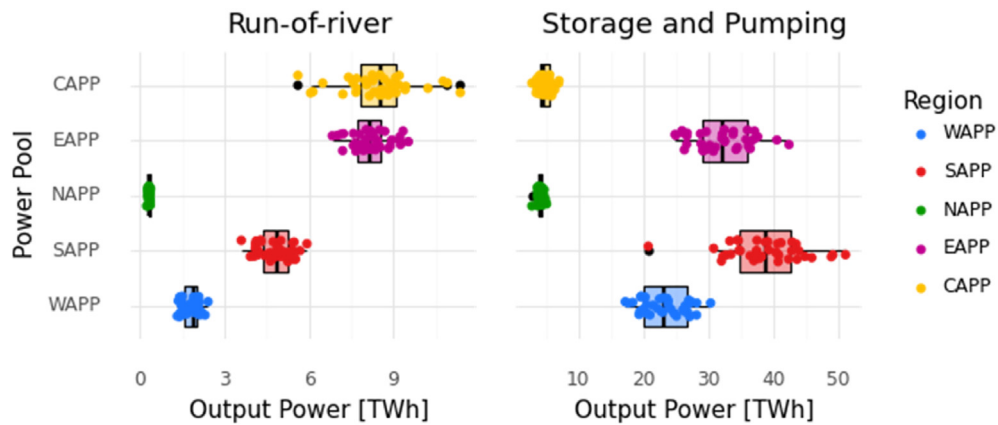


Fig. 10. Hydropower generation for run-of-river and storage/pumping for all power pools and climate years. The box shows the range between the 25th and the 75th percentile of the generation for all the climate years considered.

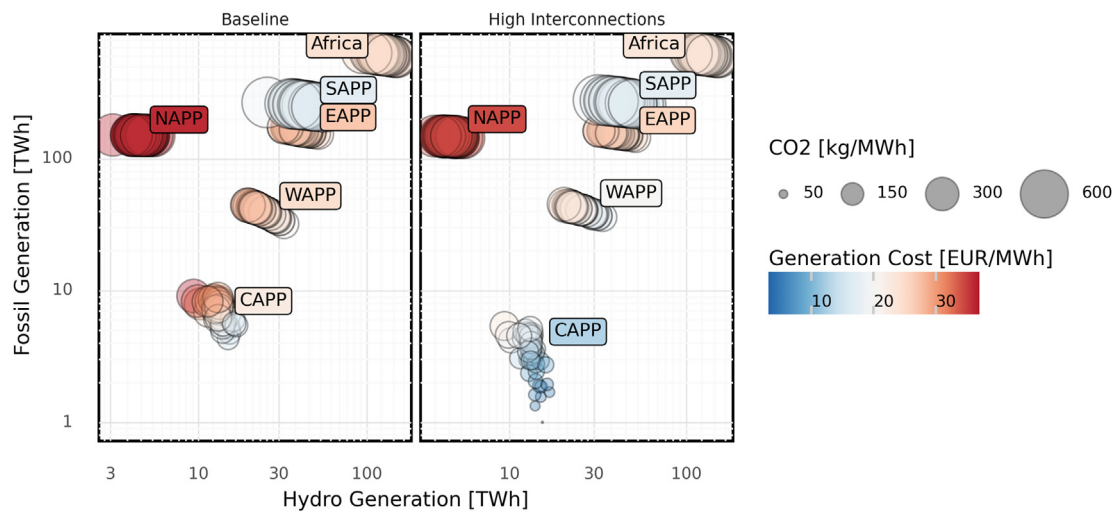


Fig. 11. Generation from hydro and fossil units in both scenarios and for all climate years and all power pools. The size and the colour of the bubbles show the fuel cost and the carbon intensity for the specific climate year. Axes are presented in logarithmic scale in order to increase the readability of the plot. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

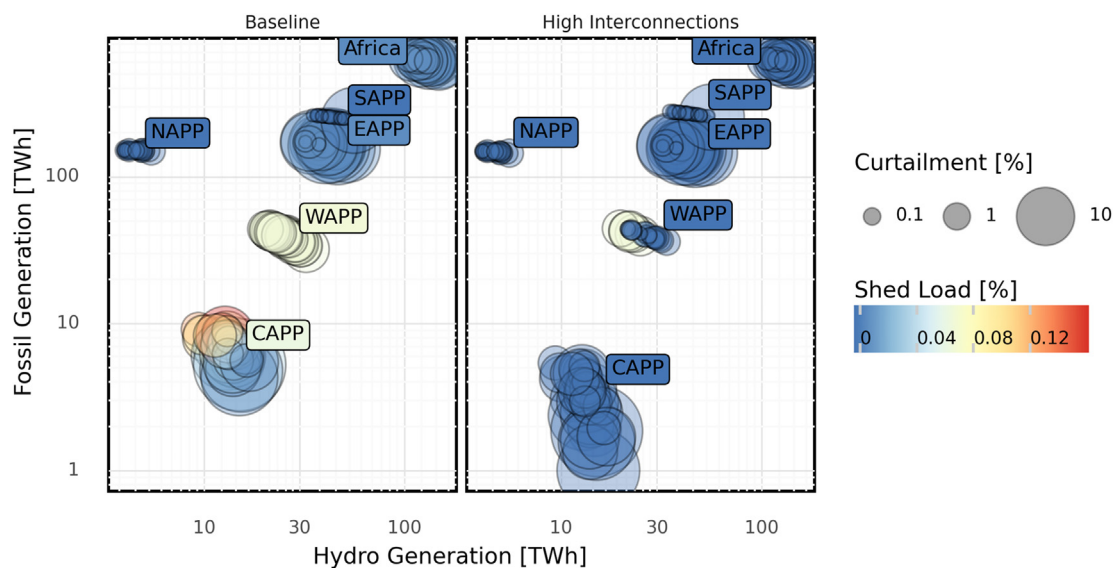


Fig. 12. Generation from hydro and fossil units in both scenarios and for all climate years and all power pools.

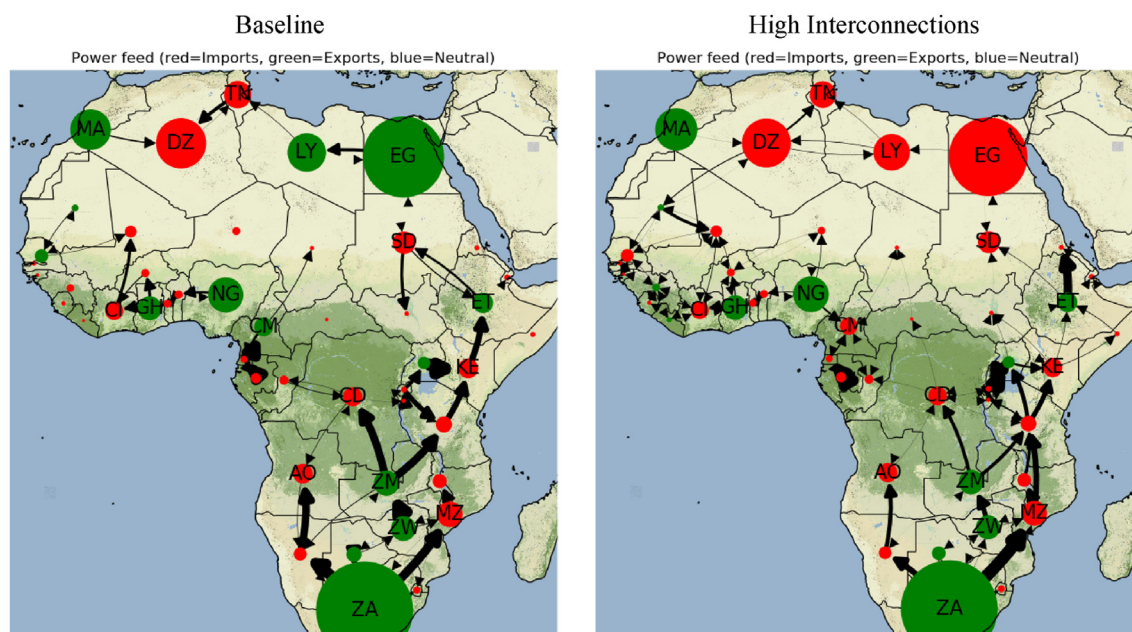


Fig. 13. Power feed in African countries in both scenarios for the calibrated climate year (2018). The bubble size indicates the total annual demand. Colour code indicates that a particular country is either importing (red) or exporting (green) electricity to its neighbours. Size of the arrows indicates the amount of energy being transferred. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

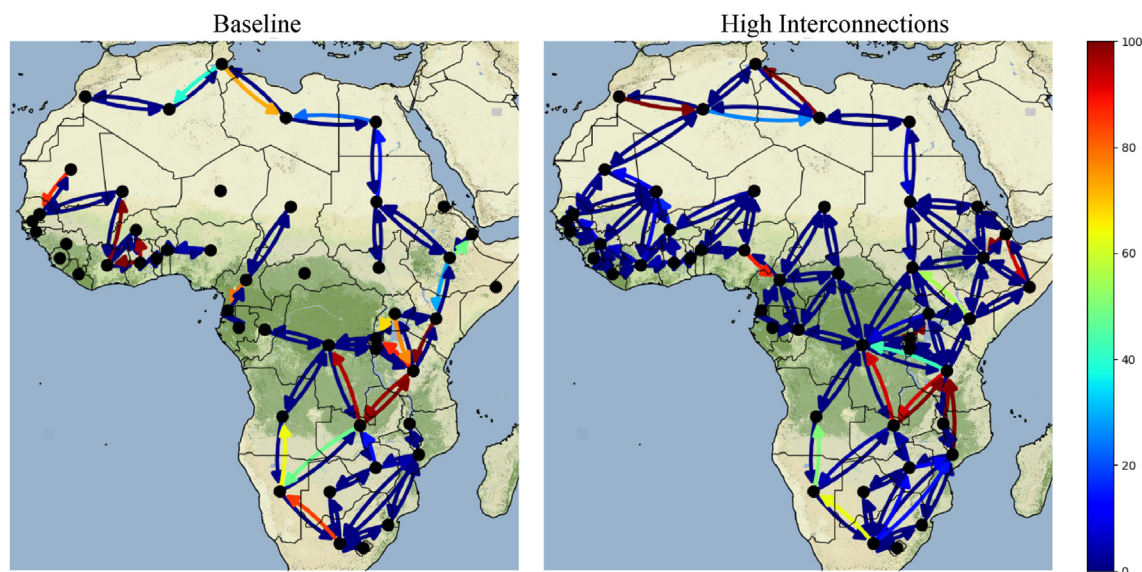


Fig. 14. Congestion in the cross-border interconnection lines. The colour gradient indicates the proportion of congested hours in the year. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

SAPP (93%) are reduced, while the newly established link between CAPP and WAPP is highly congested (96%).

4.5. Water stress

4.5.1. Water withdrawal and consumption

An important impact of electricity generation is the freshwater consumption linked to the operation of power plants. The annual need of fresh water (excluding sea and polluted water) for cooling purposes (gas, oil, nuclear, coal, biomass, waste, peat etc.) is computed for the considered climate years according to the methodology presented earlier in the paper. A summary of water

consumption (WC) and water withdrawals (WW) for thermal power plants, in relation to the average annual inflows and generation costs is shown in Fig. 16. The difference between different power pools is due to different energy mixes and to different weather conditions. As expected, power pools with the highest variability (i.e. the difference between the minimum and maximum use of freshwater) are the regions with highest shares of renewables (wind, solar and hydro). The highest average WC of 1.13 m³/MWh of total electricity generation is observed in SAPP because of harsh climate conditions, low water availability, high total demand and most water intensive technology mix, and the lowest of 0.006 m³/MWh in CAPP. The highest average WW of 43.54 m³/MWh is

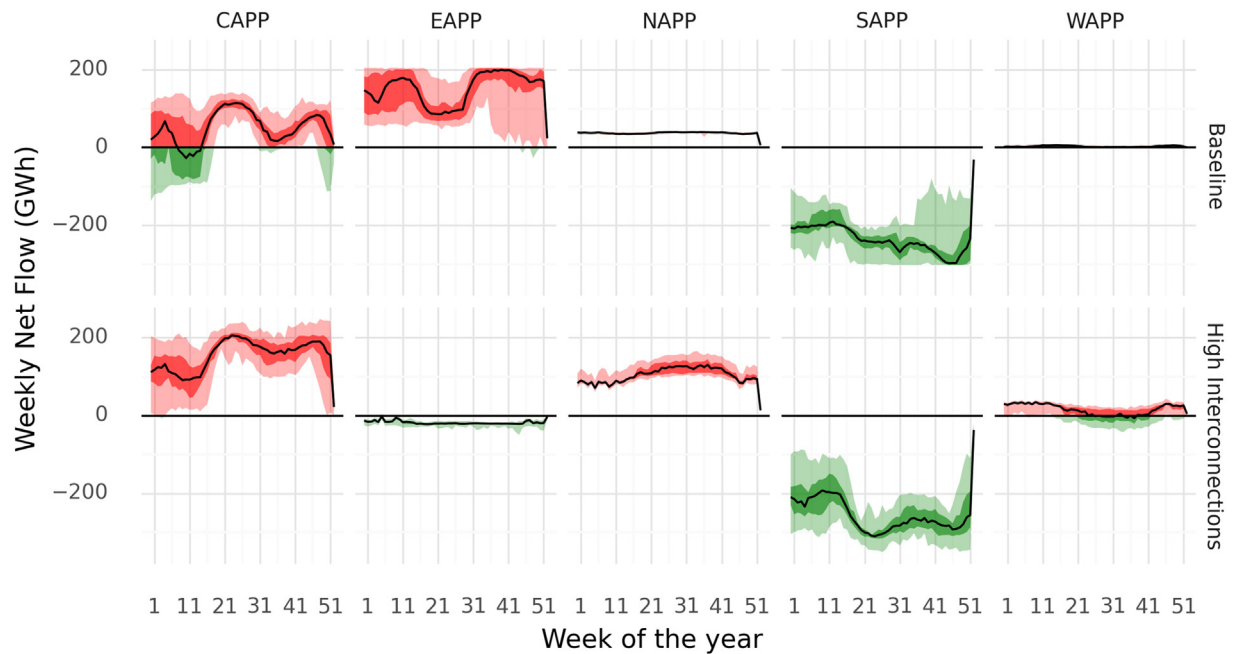


Fig. 15. Average weekly net power flow between power pools. Colour transparency indicates 25th and 75th percentiles in different climate years. Red indicates net importing and green indicates net exporting power pools. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

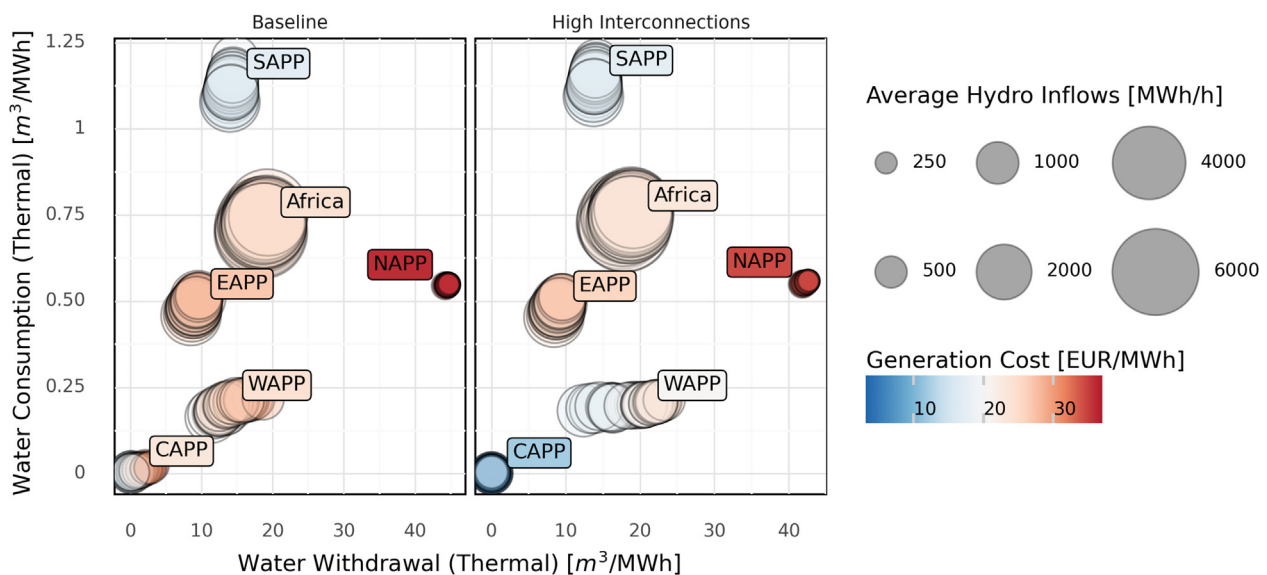


Fig. 16. Water withdrawal and water consumption for thermal generation purposes in both scenarios, for all climate years and all power pools (colour coded). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

observed in NAPP because of largest share of once through cooling systems and the lowest of $0.55 \text{ m}^3/\text{MWh}$ in CAPP. The average WC on continental scale ranges between 0.69 and $0.77 \text{ m}^3/\text{MWh}$, while WW range between 17.52 and $19.29 \text{ m}^3/\text{MWh}$. When compared to values computed for the European power pools [64] this is relatively low.

4.5.2. Absolute water exploitation index

As previously defined, absolute WEI stand for all WC and WW used for power generation (including thermal generation and water evaporation in hydro reservoirs). A proper calculation of absolute WEI is challenging, because of the lack of proper data or large

number of external variables impacting the consumption. For example, multipurpose dams (which are quite common in Africa) are used for power generation, irrigation, tourism and recreation, drinking water supply, flood control, navigation etc. Water levels in those reservoirs must therefore be kept within certain non-energy related bounds. Water evaporation caused by those activities should also subtracted when calculating WEI, as proposed in Ref. [65]. This is also further clarified in Ref. [66] where water consumption allocation factors of hydropower units based on the number of economic benefits are defined for different power capacity ranges. The same principles were also applied for calculation of absolute WEI in this study. A summary of the computed WEI is

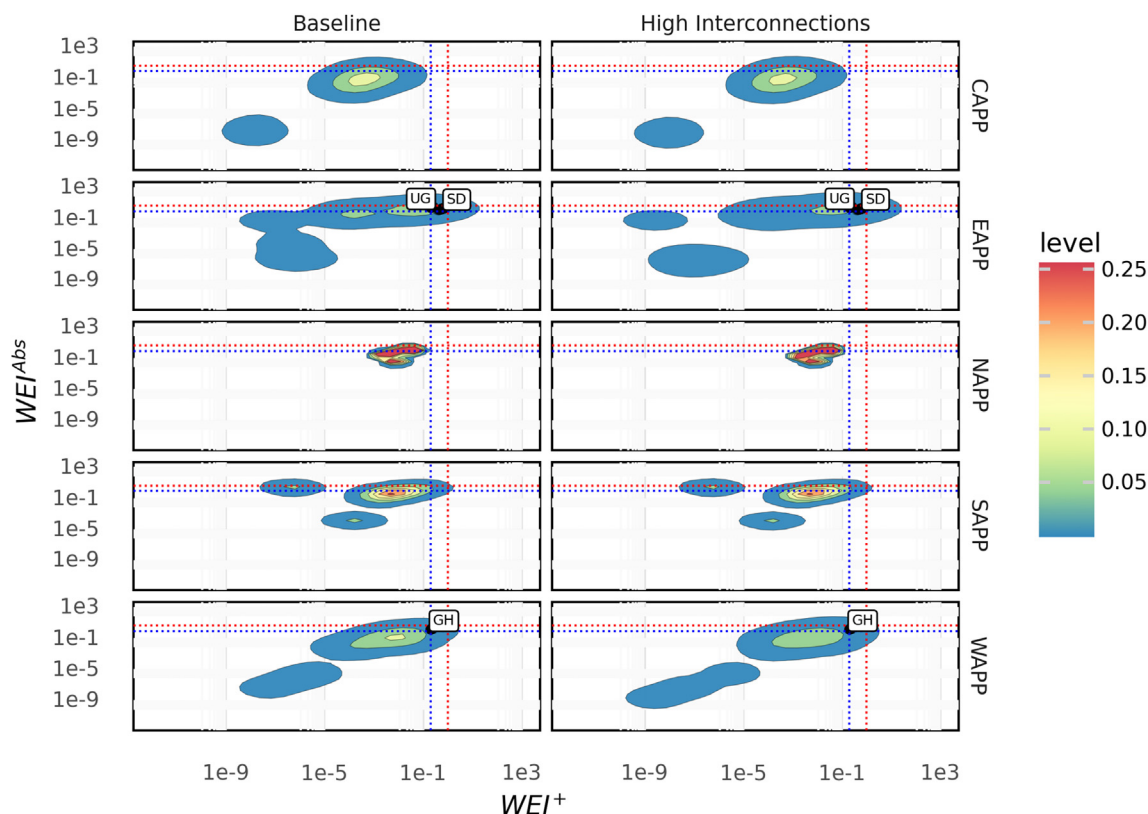


Fig. 17. 2D probability density function of the water exploitation indexes for the 48 African countries and for the 39 climate years. Blue lines indicate acceptable WEI values (0.2 for WC and 1 for WW) and red lines indicate critical WEI values (1 for WC and 5 for WW). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

presented in Fig. 17. Dependency on external freshwater resources is particularly visible in East Africa, where most countries, including Egypt and Sudan, largely depend on the fresh water from the Nile River. These two countries are consuming more water then locally available (on average nearly 5.2 times as much in Egypt and 6.4 in Sudan) and withdrawing above critical threshold of 5 withdrawals of the same water resource (131.9 times in Egypt and 20.9 times in Sudan). The only other countries withdrawing locally available freshwater resources above the critical threshold are Morocco and Libya from NAPP, with an average of 52.3 and 49.1 times higher than the threshold respectively. In terms of total availability of freshwater resources, Sudan has the highest average WEI^+ of 0.52 (ranging between 0.39 and 0.88). The only other African country consuming more fresh water than the proposed threshold of 0.2 [40] is Ghana in WAPP. Although average WEI^+ index of Ghana (0.17) is below the threshold, a maximum WEI^+ of 0.25 can be reached in certain climate years. As expected, the difference between both scenarios is low, the largest impact is observed in WAPP and CAPP, the only two power pools with limited interconnections between the member states. The difference between the two scenarios, although marginal, can still be observed. In EAPP, grid expansion positively impacts water consumption. This can be observed in countries with larger water resources such as Tanzania, Ethiopia and Rwanda. In WAPP, new grid infrastructure utilizes untapped water potential between southern and previously unconnected countries such as Liberia, Sierra Leone and Guinea. This enables a more efficient operation of the existing power plant portfolio and reduces overall WC in those countries by 12, 31 and 9% respectively. However, new interconnections are not sufficient to limit the overexploitation in Ghana, where WC in the High Interconnections scenario increases slightly by 1.3%. WEI^+ and WEI^{Abs}

in other power pools are on the same level as in the Baseline scenario.

4.5.3. Thermal only water exploitation index

The calculation of the WEI of thermal units can be more representative because water resources are used only for power generation. The correlation between WEI^{Abs} and WEI^+ in terms of freshwater resources is presented in Fig. 18. Similar patterns as before are observed, however all countries are within consumption bounds. Excessive water withdrawals in terms of internal freshwater availability is only observed in Morocco (on average same water is withdrawn 7.3 times). There are three other countries whose withdrawals are still above the proposed WW threshold. Libya in NAPP (on average water is withdrawn 113 times), Egypt in EAPP (on average water is withdrawn 1.99 times) and Malawi in SAPP (on average water is withdrawn 3.2 times). In terms of total (internal and external) freshwater resources Malawi and Libya, two countries depending on internal freshwater resources only, are the only countries above the proposed WW threshold (3.2 and 1.13 times respectively). Compared to the absolute WEI, difference between scenarios in WEI of thermal only units is more pronounced. For example, in Libya WW in High interconnections scenario are reduced by 11.8% (down to 1.01), while in most CAPP countries WW fell significantly (as much as 75% in Chad and Central African Republic). This reduction can be directly contributed to the reduced operation of the local (expensive and mostly OIL fired) backup thermal units, due to increased inflows from the SAPP. As the total annual demand between SAPP and CAPP differs by two orders of magnitude, increase of WW and WC in SAPP countries is negligible.

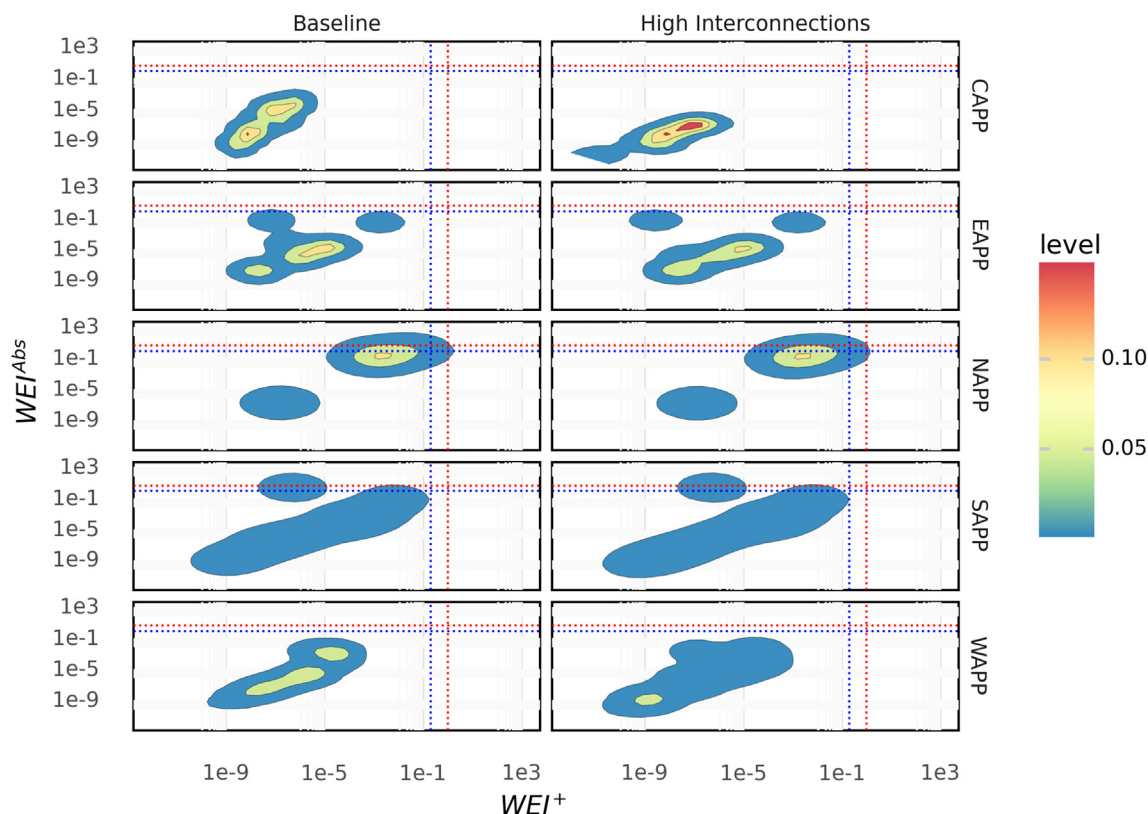


Fig. 18. 2D probability density function of the water exploitation indexes for thermal units only and for the 48 African countries and 39 climate years. Blue lines indicate acceptable WEI values (0.2 for WC and 1 for WW) and red lines indicate critical WEI values (1 for WC and 5 for WW). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

5. Conclusion

In this study, a detailed power system modelling framework is presented for the African continent and applied to assess the induced water stress. The proposed model is an open source unit commitment and power dispatch model with one node per African country. All large-scale operating generation units are reviewed and parametrized using different data sources. The modelling framework is then calibrated with historical data from 2018. Two scenarios are defined to assess the effects of the short-term expansion of the African power grid in the five African power pools. Insights regarding the importance of weather variability are obtained simulating 39 different annual climatic conditions (climate years).

This paper aims to answer two research questions:

- to examine current and future electricity situation and the potential of power trade between countries in the selected power pools
- to investigate the relationship between the water and energy with the main focus on the role of hydropower

The results indicate that the system adequacy is insufficient in many countries: significant amounts of load shedding are required in several isolated or poorly interconnected countries in the five power pools (mostly in WAPP and CAPP). Interestingly, increasing the interconnection capacities reveals to be an effective solution, bringing the load shedding to almost zero. Increased interconnections also reduce both VRES curtailment, thus increasing the overall flexibility of the system, and the marginal prices. Congestion in several major cross-border lines is however observed

in all scenarios, because of the limited transfer capacities from North to South, East to West and vice versa.

The analysis of the water-energy nexus indicators reveals that the water stress induced by power generation activities is problematic in some power pools. The influence of water temperatures on water availability for power plant operation is not considered. However, the dispatch of traditional generation is influenced by their profitability, as computed by the optimal dispatch model. Water consumption from power plants exceeds 0.2 times the water availability in several countries. A particularly vulnerable country is Sudan, with water withdrawals up to 6.4 times the water availability and total water consumption of 0.52 times the water availability. Highest freshwater abstraction among all analysed climate years of 160.1 m³/MWh is recorded in Sudan, a country with excessive dependency on external freshwater resources from the Nile River. Despite this, the current exploitation of water resources for power generation remains acceptable in the majority of African countries. It is important to note that Africa is currently undergoing a rapid growth in electricity demand, which will most likely impact the previous conclusion.

In future works, the model will be used as a reference model to assess various scenarios regarding the further development of African energy systems and their environmental impacts. For example, advanced analyses of increased demands levels together with the further deployment the grid and of variable renewable energy sources is made possible by the proposed model, which is released under an open license² and will therefore benefit the entire scientific community.

² <https://doi.org/10.5281/zenodo.4321628>.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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The views expressed are purely those of the authors and may not in any circumstances be regarded as stating an official position of the European Commission.

A.2. Demand balance

The main constraint to be met is the power supply-demand balance, for each period and each zone, in the day-ahead market as proposed in the following equation:

According to this restriction, the sum of the power generated by all the units present in the node (including the power generated by the storage units), the power injected from neighbouring nodes, and the curtailed power from intermittent sources is equal to the day ahead load in that node and power consumed for energy storage, minus the shed load.

A.3. Hydro-thermal mid-term scheduling

$$\text{MinTotalSystemCost} = \sum_{\forall u,i} \left(\begin{aligned} & \text{CostStartUp}_{i,u} + \text{CostShutDown}_{i,u} + \\ & \text{CostFixed}_u \cdot \text{Committed}_{i,u} + \\ & \text{CostVariable}_{i,u} \cdot \text{Power}_{i,u} + \\ & \text{CostRampUp}_{i,u} + \text{CostRampDown}_{i,u} + \\ & \text{PriceTransmission}_{i,l} \cdot \text{Flow}_{i,l} + \\ & \sum_n \left(\text{CostLoadShedding}_{i,n} \cdot \text{ShedLoad}_{i,n} \right) + \\ & \text{VOLL}_{\text{Power}} \cdot (\text{LL}_{\text{MaxPower},i,n} + \text{LL}_{\text{MinPower},i,n}) + \\ & \text{VOLL}_{\text{Reserve}} \cdot (\text{LL}_{2U,i,n} + \text{LL}_{2D,i,n} + \text{LL}_{3U,i,n}) + \\ & \text{VOLL}_{\text{Ramp}} \cdot (\text{LL}_{\text{RampUp},u,i} + \text{LL}_{\text{RampDown},u,i}) \end{aligned} \right) \quad (6)$$

ANNEX A

In the following sections, a brief summary of submodules and most important equations from the Dispa-SET model, relevant for this study, is presented and discussed in more details.

A.1. Objective function

The objective function of the Dispa-SET model can be summarized as follows:

$$\begin{aligned} \sum_u (\text{Power}_{u,i} \cdot \text{Location}_{u,n}) + \sum_l (\text{Flow}_{l,i} \cdot \text{LineNode}_{l,n}) = \text{Demand}_{DA,n,h} + \sum_r (\text{StorageInput}_{s,h} \cdot \text{Location}_{s,n}) - \text{ShedLoad}_{n,i} - \text{LL}_{\text{MaxPower},i} \\ + \text{LL}_{\text{MinPower},i} \end{aligned} \quad (7)$$

Here *TotalSystemCost*, in EUR, are defined as the sum of different cost items such as: start-up and shut-down, fixed, variable, ramping, transmission-related and load shedding (voluntary and involuntary) over the optimization period.

Dispa-SET MTS is a submodule of the Dispa-SET library. It provides a relaxed (LP) version of the classical and highly detailed MILP formulation of the unit commitment and power dispatch problem. Constraints linked to power plant cycling such as start-up and shut-down time, ramping rates, minimum up and down time, must run power and unit commitment are neglected. This enables a more efficient computation of the optimal (instead of exogenously supplied) reservoir levels which are then used as minimum level constraints in the UCM module. In the latter, the problem is split into smaller optimization problems that are run recursively throughout the year. The initial values of the optimization for any given day are the final values of the optimization of the previous

day. A look-ahead period is included and then discarded to avoid issues linked to the end of the optimization period such as starting low cost but non-flexible power plants. This optimal hydro allocation is particularly relevant for the analysis of the water-energy nexus, where availability of water resources for energy needs is crucial for the operation of the power generators.

ANNEX B

B.1. Fuel price forecasts

Variability of fuel prices is estimated based on a fuel price estimation algorithm. The proposed algorithm maps large amounts of information, such as geography, local production and transportation and resource availability to a single fuel price. For each fuel type, one decision variable performs each of the three categories is used to either increase or decrease the final fuel price, as proposed in Ref. [67]. In each zone, decision variables are defined by the: local fuel production (yes/no) and means of transportation (rail, truck and/or ship or through international pipelines), access to a major sea port (yes/no), and local resource availability (biomass, and peat scarcity/availability). A summary of the proposed algorithm is presented in Fig. 19.

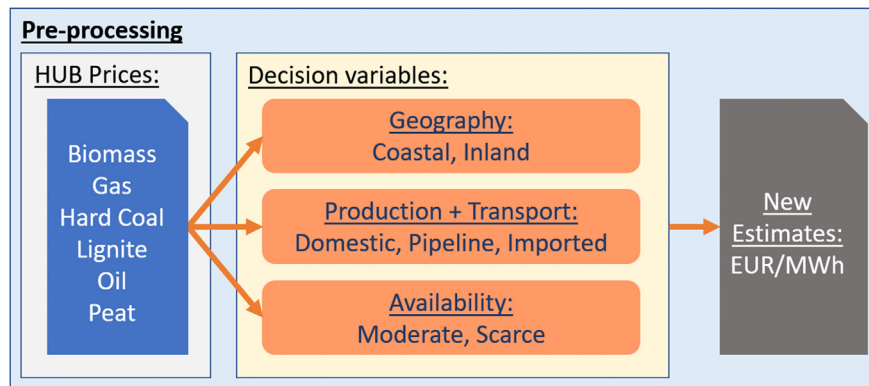


Fig. 19. Fuel price estimation algorithm based on three decision variables geography, fuel production and availability.

NOMENCLATURE

Abbreviations Description Unit

AF	Availability factor [–]
AIR	Air cooling [–]
BIO	Biomass and biogas [–]
CHP	Combined heat and power [–]
COMC	Combined cycle [–]
Ext	External [–]
GAS	Gas [–]
GEO	Geothermal [–]
GTUR	Gas turbine [–]
HDAM	Hydro dam (turbine) [–]
HPHS	Pumped hydro (pumped/turbine) [–]
HRD	Hard coal [–]
HROR	Hydro run-of-river [–]
ICEN	Internal combustion engine [–]
LIG	Lignite [–]
LL	Lost Load [–]
MDT	Mechanical or natural draft cooling tower [–]
MILP	Mixed integer linear programming [–]
MTS	Mid-term scheduling [–]
NTC	Net transfer capacity [–]
NUC	Nuclear [–]
OIL	Oil [–]
OTF	Once-through cooling system (fresh water) [–]
OTH	Other energy carriers including electric vehicles [–]

OTS	Once-through cooling system (saline water) [–]
PHOT	Photovoltaics [–]
RES	Renewable energy sources [–]
SCSP	Concentrated solar power [–]
STUR	Steam turbine [–]
SUN	Solar [–]
TSO	Transmission System Operator [–]
UCM	Unit-commitment and power dispatch models [–]
VRES	Variable energy sources [–]
WA	Water availability [m ³]
WAT	Hydro [–]
WC	Water consumption [m ³]
WEI	Water exploitation index [–]
WIN	Wind [–]
WW	Water withdrawal [m ³]

Sets Description Unit

i	Time step in the current optimization horizon [–]
l	Transmission lines between nodes [–]
n	Zones [–]
$p2h$	P2HT units [–]
u	Units [–]

Parameters Description Unit

$CostFixed_u$	Fixed costs [EUR/h]
$CostLoadShedding_{i,n}$	Shedding costs [EUR/MWh]
$CostRampDown_{i,u}$	Ramp-down costs [EUR/MW]
$CostRampUp_{i,u}$	Ramp-up costs [EUR/MW]
$CostShutDown_{i,u}$	Shut-down costs for one unit [EUR/u]
$CostStartUp_{i,u}$	Start-up costs for one unit [EUR/u]
$CostVariable_{i,u}$	Variable costs [EUR/MWh]

$Demand_{DA,n,h}$ Hourly demand in each zone [MW]
 $LineNode_{l,n}$ Line-zone incidence matrix $\{-1,+1\}$ [–]
 $LoadShedding_n$ Maximum value of load shedding [MW]
 $Location_{u,n}$ Location {binary: 1 u located in n} [–]
 $PriceTransmission_{i,l}$ Price of transmission between zones [EUR/MWh]
 $StorageInput_{s,h}$ Charging input for storage units [–]
 $Technology_u$ Technology type [–]
 $VOLL_{Power}$ Value of lost load due to power output [EUR/MWh]
 $VOLL_{Ramp}$ Value of lost load due to ramping [EUR/MWh]
 $VOLL_{Reserve}$ Value of lost load due to lack of reserve capacities [EUR/MWh]
Variables Description Unit
 $Flow_{i,l}$ Flow through lines [MW]
 $LL_{2D,i,n}$ Deficit in reserve down [MW]
 $LL_{2U,i,n}$ Deficit in reserve up [MW]
 $LL_{3U,i,n}$ Deficit in reserve up - non spinning [MW]
 $LL_{MaxPower,i,n}$ Deficit in terms of maximum power [MW]
 $LL_{MinPower,i,n}$ Power exceeding the demand [MW]
 $LL_{RampDown,u,i}$ Deficit in terms of ramping down for each plant [MW]
 $LL_{RampUp,u,i}$ Deficit in terms of ramping up for each plant [MW]
 $Power_{i,u}$ Power output [MW]
 $ShedLoad_{i,n}$ Shed load [MW]
 $TotalSystemCost$ Total system cost [EUR]

Integer variables Description Unit

$Committed_{i,u}$ Committed status of unit at hour h {1 0} or integer [–]

Credit author statement

Matija Pavičević: Conceptualization, Methodology, Model Development, Data curation, Writing & Original draft preparation, Visualization, Investigation, Reviewing and Editing. Matteo De Felice: Data curation, Model Development, Visualization, Reviewing and Editing. Sebastian Busch: Data curation, Supervision, Reviewing. Ignacio Hidalgo González: Data curation, Supervision, Reviewing. Sylvain Quoilin: Conceptualization, Methodology, Model Development, Supervision, Reviewing and Editing.

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